

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8628**

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 409	Seat No: 2022700006
Course Section: 1		

L32_619_986_1_1505_41460



Name of the candidate. Rakshita Borkar

Candidate's UID number:

2	0	2	2	7	0	0	0	0	6
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Examination class room number:

4	0	9
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Invigilator's signature with date:

 08/10/25

(To be filled in by the candidate)

EXAMINATION: MSE BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: CSE DS

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 08/10/2025 Wednesday

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q1a

$$f(w) = \frac{1}{2} \|xw - y\|^2.$$

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~~Q.20~~

Q.1. b

Overfitting occurs when the model complexity is too high. This is because the model memorises patterns as well as noise. Using overly complex models for simple problems can lead to the model learning not just the necessary information but the noise patterns as well.

This occurs when we use overly complex model; eg using quadratic equations for a linear regression problem.

This can be identified when the training accuracy is high but the test accuracy is low.

To overcome overfitting, we need to ~~either~~ use a simpler model ~~or~~
eg

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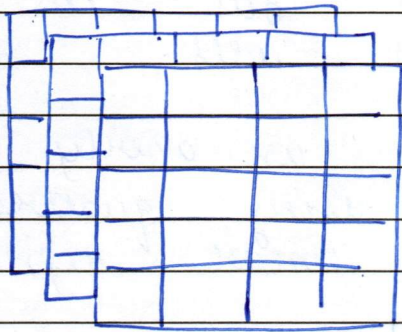
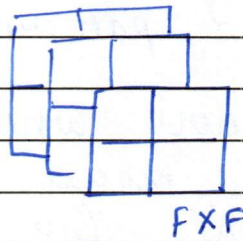
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Q.2a

For input size $H \times W \times C$
with K filters of size $F \times F$

2D convolution complexity can
be derived as

 $H \times W \times C$  $F \times F$ K filters.

Number of multiplications
will be

$$= (W - F + 1) * (H - F + 1) * (F * F)$$

[illegible]



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Q.2b

Linear undercomplete autoencoder is the type of autoencoder where the bottleneck has fewer neurons than the input neurons.

Undercomplete autoencoders are used for dimensionality reduction

✶ An overcomplete autoencoder on the other hand is the opposite of an undercomplete autoencoder.

An overcomplete autoencoder has more neurons in the bottleneck or the latent vector than its input layer.

Without regularization, an overcomplete autoencoder does not capture might ~~any relevant~~ features and up capturing noise. An overcomplete autoencoder without regularization is practically just copy pasting values from input to output. Hence, it can and does tend to learn the identity function where all the weights are and there is no change in the inputs to outputs. Hence, we use regularization to prevent combat this from happening.

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Q-3a

$$\frac{\partial L}{\partial h_{t-k}}$$

Gradient descent is calculated as .

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial h_{t-k}}$$

The loss function of an RNN would be $|\hat{y} - y|^2$

An RNN is ~~a sequential~~ a neural network which memorises the previous inputs . It is usually used on sequential data like time series values .

It feeds back the output to the ~~training~~ hidden layer again .

~~An LS~~ However, an RNN cannot remember data from very old values . If the model needs to refer to previous data RNNs do not have the ability to do that i.e they do not have "long term" memory . And as we keep giving more inputs the model starts forgetting previous data . Hence LSTMs were introduced . LSTM stands for Long Short Term Memory . LSTMs have two states - Cell State & Hidden State . LSTMs have 3 gates Input gate , forget gate and output gate . The forget gate tells the model what information it needs to memorise and what it needs to forget . The forget gate

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Q36

Gating equations of a GRU are.

update gate.

$$\cancel{w_z} \quad z = \cancel{w_z} \sigma(w_{h-t}, x_t + b_z)$$

Reset gate.

$$r = \sigma(w_{h-t}, x_t + b_r)$$

Update equation.

$$\cancel{(1 - z_{t-1}) h_{t-1} + r z_{t-1}}$$

$$(1 - z_{t-1}) h_{t-1} + r z_{t-1}$$

GRUs have fewer parameters than LSTMs because GRUs have fewer gates. The forget and input gates in LSTMs are combined in GRUs. That is the cell state & hidden state are combined in GRUs. LSTMs have input gate, forget gate and output gates whereas GRU has only 2 gates input & reset gate and update gate. GRUs were made with the intention of making calculations more efficient. LSTMs are computationally intensive and having fewer gates and fewer parameters makes GRUs have fewer parameters. This would improve generalization in real world as more & more problems can be ~~solved~~ solved with GRUs which ~~are~~ ~~if~~ require efficiency.



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❖ In some real world or rather most real world problems require that efficiency is prioritized over extreme accuracy. Accuracy is important as well but can be ~~looked~~ overlooked if a more efficient algorithm exists. ORU Hence ~~it~~ can be more efficient and used in a wider range of real-world applications.

[illegible]

[illegible]

[illegible]